

DEEPAK ARAVINDAN

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PROFESSIONAL SUMMARY

AI Product Engineer with 6+ years of combined experience building and shipping AI-powered products end-to-end — from defining product vision to writing production code. Hands-on with LLM orchestration, AI agent architectures, RAG pipelines, computer vision, and generative AI, backed by deep full-stack engineering skills (Python, React/Node.js, AWS, Docker). Proven ability to prototype rapidly, ship AI features that drive measurable user outcomes (30–40% faster deployments, 15–22% adoption lifts), and collaborate across ML engineering, design, and business teams. Former startup builder with the product instinct, technical depth, and ownership mindset that frontier AI companies demand.

TECHNICAL SKILLS

AI & ML:	LLM Orchestration (GPT-4, Claude, LLaMA, Mistral), AI Agents & Tool Use, RAG Architecture, Prompt Engineering & Optimization, Fine-Tuning (LoRA), Computer Vision, NLP, Model Evaluation, Hallucination Reduction, Guardrails & Safety
Languages & Frameworks:	Python, JavaScript/TypeScript, React, Node.js, Express, REST APIs, GraphQL, Kafka, SQL
Infrastructure & MLOps:	AWS (Lambda, EC2, S3, SageMaker), Docker, CI/CD Pipelines, FFmpeg, GStreamer, Microservices, Edge AI Deployment, Model Serving
Data & AI Tooling:	LangChain, LlamaIndex, Vector Databases (Weaviate, Pinecone), OpenAI API, Embeddings, Mixpanel, Google Analytics, A/B Testing, Data Pipelines
Product & Design:	PRDs, Roadmapping, User Stories, RICE/MoSCoW Prioritization, Figma, Miro, Agile/Scrum, User Research, Competitive Analysis
Tools:	Jira, Git, GitHub Actions, Postman, Jupyter Notebooks

PROFESSIONAL EXPERIENCE

AI Product Engineer — Variphi

Jan 2025 – Present

- Architected and shipped a multi-agent AI system for real-time safety-violation monitoring, orchestrating LLM-based agents with tool-use capabilities (LangChain, OpenAI API) for autonomous image analysis and alert generation — reduced client deployment time by 30%.
- Built end-to-end generative AI pipeline for on-demand image synthesis, implementing prompt optimization, guardrails for content safety, and evaluation frameworks to ensure output quality at production scale.
- Designed and optimized HALO Accelerator — a zero-touch provisioning system (Python, Docker, AWS Lambda) — cutting enterprise onboarding from weeks to days (40% reduction) and enabling self-serve configuration for new clients.
- Engineered NVR-based edge AI monitoring solution integrating computer vision models with FFmpeg & GStreamer pipelines for local inference, reducing cloud costs and lowering system crashes by 25%.
- Led R&D; evaluation of foundation models (GPT-4, Claude, open-source LLMs) across generative and perceptive AI use-cases; designed RAG pipelines with vector databases for knowledge retrieval, boosting pilot adoption by 15%.
- Owned product roadmap and technical PRDs for all AI features; collaborated with ML engineers on model selection, prompt engineering, fine-tuning strategies, and production deployment.

Product Engineer — ValueMatrix.ai

Feb 2023 – Dec 2024

- Owned end-to-end roadmap and hands-on development for a candidate evaluation platform used by enterprise clients — defined PRDs, wrote user stories, and personally shipped core features, improving client retention by 22%.
- Led a small engineering team to build LinkedIn integration for real-time candidate scoring and a Kafka-based analytics pipeline, reducing manual tasks by 60% and cutting API latency by 30%.
- Designed and launched Big5 psychometric assessment module end-to-end — from product spec to frontend implementation to release — driving 30% increase in upsell revenue.
- Established security best practices across the MERN stack: built middleware for input validation and XSS prevention, lowering vulnerabilities by 40% and improving platform reliability.
- Conducted extensive user research and competitive analysis, informing two high-ROI feature releases; streamlined PRD and user story workflows, reducing feature delivery cycle by 18% and boosting UX task completion by 25%.
- Built Mixpanel & GA dashboards with custom event tracking; iterated UI based on data insights and ran A/B tests on activation flows, lifting activation rates by 19% within four weeks.

Frontend Engineer & IoT Developer — Autonomo Technologies

Apr 2021 – Jan 2023

- Architected and shipped React-based admin portal for autonomous store operations powered by computer vision checkout — reduced operational delays by 35%.
- Built "Hoody" — a real-time fault-tracking system for CV-based checkout hardware, integrating IoT sensor telemetry for predictive diagnostics; reduced support escalations by 40%.

- Integrated IoT peripherals (sensors, dispensers) with cloud backend, enabling automated device management and laying groundwork for AI-driven predictive maintenance.

Software Engineer — Smartlink / DIGISOL

Earlier Role

- Developed scalable router management platform handling 50K+ devices with 99.9% uptime; implemented security hardening and performance optimizations for production infrastructure.

AI PROJECTS & SIDE BUILDS

- **RAG Knowledge Assistant:** Built an internal documentation search tool using LangChain, Weaviate vector database, and OpenAI embeddings — reduced information retrieval time by 50% across engineering team.
- **AI Resume Screener:** Developed end-to-end candidate-job matching prototype using fine-tuned transformer models and prompt engineering — achieved 92% accuracy on benchmark dataset.
- **Multi-Agent Task Automation:** Designed autonomous agent workflow using LangChain agents with tool-use (web search, code execution, API calls) for automated research and reporting tasks.
- **MLOps Pipeline:** Set up model versioning, automated evaluation, and CI/CD deployment pipeline on AWS SageMaker for continuous model improvement and A/B model serving.

CERTIFICATIONS

- Certified Scrum Product Owner (CSPO)
- Generative AI with Large Language Models — DeepLearning.AI & AWS
- Google Analytics Individual Qualification (GAIQ)
- Google UX/UI Design Professional Certificate

EDUCATION

B.Tech, Computer Science — SRM University